

# E-COMMERCE SALES DATA ANALYSIS PROJECT REPORT

## 1. Project Overview

This project demonstrates end-to-end data analysis using Excel, MySQL, and Power BI. The goal was to convert raw e-commerce sales data into meaningful business insights through structured data cleaning, SQL-based analysis, and interactive dashboard reporting.

## 2. Data Cleaning & Preparation (Excel)

- Removed duplicates and handled missing/null values.
- Used TRIM function to remove extra spaces By Power Query.
- Standardized text using Power Query.
- Corrected data types (Date, Numeric, Etc).
- Created additional calculated columns where required.
- Performed Pivot Table analysis (Region, Category, Product).

## 3. SQL Business Analysis (MySQL)

- Total Sales, Profit, and Orders calculation.

|   | TOTAL_ORDER | TOTAL_SALES | TOTAL_PROFIT |
|---|-------------|-------------|--------------|
| ▶ | 15000       | 33200489.83 | 7161303.79   |

- Region-wise Sales & Profit analysis.

|   | REGION | TOTAL_ORDER | TOTAL_SALES | TOTAL_PROFIT |
|---|--------|-------------|-------------|--------------|
| ▶ | North  | 3803        | 8404625.63  | 1822643.96   |
|   | West   | 3688        | 8389716.58  | 1821011.44   |
|   | East   | 3778        | 8229929.20  | 1765469.08   |
|   | South  | 3731        | 8176218.42  | 1752179.31   |

- Category & Sub-category profitability analysis.

|   | CATEGORY    | TOTAL_ORDER | TOTAL_SALES | TOTAL_PROFIT |
|---|-------------|-------------|-------------|--------------|
| ▶ | Fashion     | 5015        | 11274374.33 | 2440850.19   |
|   | Electronics | 5006        | 11114104.67 | 2395133.63   |
|   | Home        | 4979        | 10812010.83 | 2325319.97   |

|   | SUB_CATEGORY | TOTAL_ORDER | TOTAL_SALES | TOTAL_PROFIT |
|---|--------------|-------------|-------------|--------------|
| ▶ | Laptops      | 2587        | 5746160.87  | 1251794.68   |
|   | Men          | 2507        | 5669191.70  | 1219728.77   |
|   | Women        | 2508        | 5605182.63  | 1221121.42   |
|   | Furniture    | 2510        | 5420452.02  | 1172356.03   |
|   | Kitchen      | 2469        | 5391558.81  | 1152963.94   |
|   | Mobiles      | 2419        | 5367943.80  | 1143338.95   |

- Top 5 Products by Sales.

|   | PRODUCT_NAME            | TOTAL_SALES |
|---|-------------------------|-------------|
| ▶ | Butterfly Kitchen Item  | 2798504.08  |
|   | Nilkamal Furniture Item | 2723255.95  |
|   | Ikea Furniture Item     | 2697196.07  |
|   | Prestige Kitchen Item   | 2593054.73  |
|   | Adidas Men Item         | 1962657.12  |

- Bottom 5 Loss-making Products.

|   | PRODUCT_NAME         | TOTAL_SALES |
|---|----------------------|-------------|
| ▶ | Samsung Mobiles Item | 1289740.70  |
|   | Apple Mobiles Item   | 1315276.46  |
|   | Dell Laptops Item    | 1343779.61  |
|   | Realme Mobiles Item  | 1357323.46  |
|   | Redmi Mobiles Item   | 1405603.18  |

- Discount impact on Profit analysis.

|   | DISCOUNT_LEVEL | TOTAL_PROFIT |
|---|----------------|--------------|
| ▶ | 0.0            | 1809054.24   |
|   | 5.0            | 1643357.78   |
|   | 10.0           | 1298461.88   |
|   | 15.0           | 1025022.98   |
|   | 20.0           | 753186.54    |
|   | 25.0           | 441096.97    |
|   | 30.0           | 191123.40    |

- Average Sales per Product calculation.

|   | AVG(PRODUCT_SALES) |
|---|--------------------|
| ▶ | 1844471.657222     |

- Products Above Average Sales.

|   | PRODUCT_NAME            | TOTAL_SALES |
|---|-------------------------|-------------|
| ▶ | Biba Women Item         | 1891242.68  |
|   | Butterfly Kitchen Item  | 2798504.08  |
|   | Ikea Furniture Item     | 2697196.07  |
|   | Puma Men Item           | 1866874.75  |
|   | Adidas Men Item         | 1962657.12  |
|   | H&M Women Item          | 1924843.95  |
|   | Prestige Kitchen Item   | 2593054.73  |
|   | Nilkamal Furniture Item | 2723255.95  |

- Loss-making Products Below Average Profit.

|   | PRODUCT_NAME         | TOTAL_PROFIT |
|---|----------------------|--------------|
| ▶ | Redmi Mobiles Item   | 298351.43    |
|   | Asus Laptops Item    | 309147.44    |
|   | Dell Laptops Item    | 295375.84    |
|   | Realme Mobiles Item  | 283955.48    |
|   | Zara Women Item      | 389995.94    |
|   | Apple Mobiles Item   | 281924.23    |
|   | HP Laptops Item      | 337425.93    |
|   | Nike Men Item        | 394685.83    |
|   | Lenovo Laptops Item  | 309845.47    |
|   | Samsung Mobiles Item | 279107.81    |

- Top 5 Brands by Sales.

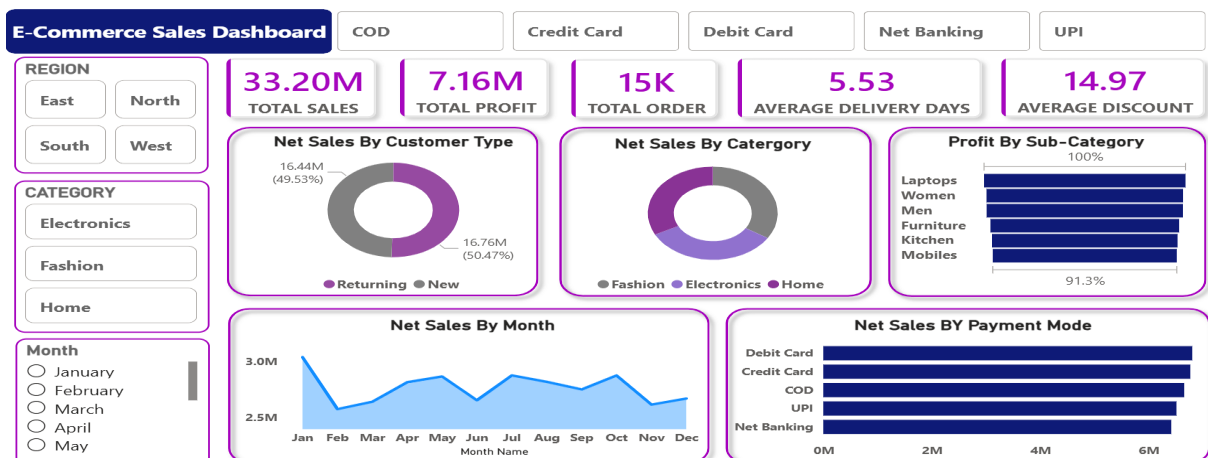
|   | BRAND     | TOTAL_SALES |
|---|-----------|-------------|
| ► | Butterfly | 2798504.08  |
|   | Nilkamal  | 2723255.95  |
|   | Ikea      | 2697196.07  |
|   | Prestige  | 2593054.73  |
|   | Adidas    | 1962657.12  |

- Top 3 Cities by Sales.

|   | CITY     | TOTAL_SALES |
|---|----------|-------------|
| ► | Mysuru   | 1910899.79  |
|   | Madurai  | 1892484.45  |
|   | Durgapur | 1868684.55  |

## 4. Power BI Dashboard Development

The cleaned data was imported from MySQL into Power BI. DAX measures were created for Total Sales, Total Profit, Average Delivery Days, and Order Count. The dashboard includes KPI cards, bar charts, product performance visuals, regional comparison charts, and slicers for dynamic filtering.



## 5. Key Business Insights

- A few products generate majority of total revenue.
- High discount percentages reduce profit margins.
- Certain regions significantly outperform others.
- Some sub-categories consistently generate losses.
- Top brands contribute heavily to overall revenue.

## 6. Recommendations & Final Opinion

Based on the analysis, the company should optimize discount strategies, focus on high-margin products, review underperforming sub-categories, and invest more in high-performing regions and brands. Future enhancements can include predictive analytics and customer segmentation.